# DT081A - Signal and Image Processing
# Exercise 5 Solutions

## Light and Spectra

**Question 1.** Explain how light is characterized by its wavelength.

**Answer:** Light is an electromagnetic wave characterized by its wavelength, which is the distance between two consecutive wave peaks. The wavelength determines the color and energy of light, with shorter wavelengths corresponding to higher energies and frequencies.

**Question 2.** What is the wavelength range that humans can perceive?

**Answer:** Humans can typically perceive light with wavelengths ranging from about 400 nm to 700 nm, which corresponds to the visible spectrum from violet to red.

**Question 3.** Calculate the frequency of light with a wavelength of 500 nm. Use the speed of light in a vacuum $c = 3 \cdot 10^8$ m/s.

**Answer:** Using the formula $f = \frac{c}{\lambda}$:

$$\begin{aligned} f &= \frac{3 \cdot 10^8 \text{ m/s}}{500 \cdot 10^{-9} \text{ m}} \\ &= 6 \cdot 10^{14} \text{ Hz} \end{aligned}$$

The frequency of light with a wavelength of 500 nm is $6 \cdot 10^{14}$ Hz.

**Question 4.** Compare and contrast a laser as a light source versus most other light sources in terms of wavelength composition.

**Answer:**

- Laser: Emits coherent light at a single wavelength or a very narrow range of wavelengths.
- Most other light sources: Emit incoherent light composed of a broad range of wavelengths.

Lasers produce highly focused, monochromatic light, while other sources typically produce diffuse, polychromatic light.

**Question 5.** Describe the process of image formation, from light reflection off surfaces to capture by an imaging device. Include the role of the reflectance function in this process.

**Answer:** The process of image formation:

1. Light source emits light

2. Light interacts with surfaces (reflection, absorption, transmission)
3. Reflected light is modulated by the surface's reflectance function
4. Reflected light enters the imaging device
5. Light is focused by the device's optics
6. Focused light is captured by a sensor
7. Sensor converts light into electrical signals

The reflectance function determines how much light is reflected at each wavelength, influencing the color and intensity of the captured image.

## Human Eye and Color Perception

**Question 6.** Describe the difference between rods and cones in the human eye. How do their roles relate to day and night vision?

**Answer:**

- Rods: More sensitive to light, responsible for night vision, cannot distinguish colors
- Cones: Less sensitive, responsible for day vision and color perception

Rods function better in low light conditions, as they are particularly sensitive to brightness, while cones require more light but provide color vision and higher visual acuity.

**Question 7.** What are the three types of cones, and what part of the color spectrum does each type respond to?

**Answer:** The three types of cones are:

- Respond to short wavelengths (blue)
- Respond to medium wavelengths (green)
- Respond to long wavelengths (red)

**Question 8.** The human eye has different sensitivity curves for red, green, and blue light. Given the sensitivity function for the green cone peaks at 550 nm, compute the corresponding frequency in Hz.

**Answer:** Using $f = \frac{c}{\lambda}$ and $c = 3 \cdot 10^8$ m/s:

$$\begin{aligned} f &= \frac{3 \cdot 10^8 \text{ m/s}}{550 \cdot 10^{-9} \text{ m}} \\ &= 5.45 \cdot 10^{14} \text{ Hz} \end{aligned}$$

The frequency corresponding to 550 nm is approximately $5.45 \cdot 10^{14}$ Hz.

**Question 9.** Define luminance and chrominance. How are these related to human vision?

**Answer:**

- Luminance: Perceived brightness of light
- Chrominance: Color information independent of luminance (i.e. hue - color tone and saturation - color purity)

Human vision is more sensitive to changes in luminance than chrominance. This property is exploited in image and video compression techniques.

## Image Formation

**Question 10.** Explain how the reflectance function of a surface influences the light captured by an imaging device.

**Answer:** The reflectance function $R$ determines the proportion of incident light reflected at each wavelength $\lambda$. It modulates the spectral power distribution of the light source, influencing the color and intensity of light captured by the imaging device. Different materials have unique reflectance functions, contributing to their perceived color and appearance in images.

**Question 11.** Given the following reflectance function for a surface:

$$R(\lambda) = \frac{1}{2}\sin\left(\frac{2\pi\lambda}{700}\right) + 0.5$$

Where $\lambda$ is the wavelength of light (in nm),

a) calculate the amount of light reflected at $\lambda = 550$ nm.

b) calculate the average reflectance over the visible spectrum (380-750 nm)

**Answer:** a) For $\lambda = 550$ nm:

$$\begin{aligned} R(550) &= \frac{1}{2}\sin\left(\frac{2\pi \cdot 550}{700}\right) + 0.5 \\ &= \frac{1}{2}\sin(\frac{11}{7}\pi) + 0.5 \\ &\approx 0.01 \end{aligned}$$

b) Average reflectance:

$$\begin{aligned} R_{avg} &= \frac{1}{750-380}\int_{380}^{750}\left(\frac{1}{2}\sin\left(\frac{2\pi\lambda}{700}\right) + 0.5\right)d\lambda \\ &= \frac{1}{370}\left[-\frac{700}{4\pi}\cos\left(\frac{2\pi\lambda}{700}\right) + 0.5\lambda\right]_{380}^{750} \\ &\approx 0.22 \end{aligned}$$

**Question 12.** An image $I(x, y)$ undergoes the following transformation:

$$I'(x, y) = aI(x, y) + b$$

Prove that this transformation preserves the relative ordering of pixel intensities.

**Answer:** Proof: Consider two pixel intensities $I_1$ and $I_2$ where $I_1 < I_2$. This means, after multiplying by $a > 0$, we get $aI_1 < aI_2$, which directly leads to $aI_1 + b < aI_2 + b$, proving the statement.

## Spectral Sensitivity and Image Capture

**Question 13.** What is the luminous-efficiency sensitivity function, and how does it relate to human vision?

**Answer:** The luminous-efficiency sensitivity function, denoted as $V(\lambda)$, describes the average sensitivity of human visual perception to light of different wavelengths. It peaks around 555 nm (green-yellow) and is used to convert radiant energy into luminous energy, accounting for the eye's varying sensitivity to different wavelengths of light in photopic (day) vision.

**Question 14.** Compare the spectral sensitivity of the human eye's red, green, and blue cones.

**Answer:**

- Red cones: Peak sensitivity around 570 nm
- Green cones: Peak sensitivity around 530 nm
- Blue cones: Peak sensitivity around 420 nm

Green and red cones have overlapping sensitivities, while blue cones are more distinct and their relative response is much lower. The combination of these sensitivities enables color perception across the visible spectrum, but with increased sensitivity in the middle of the visible spectrum.

**Question 15.** Explain how a digital camera captures color images using a Bayer filter.

**Answer:** A Bayer filter is a color filter array that allows a single image sensor to capture color images:

- It consists of a grid of red, green, and blue filters over the pixel sensors
- Each pixel captures one color component
- Demosaicing algorithms interpolate missing color information
- Results in a full-color image

**Question 16.** What is gamma correction, and why is it applied to digital images?

**Answer:** Gamma correction is a nonlinear operation applied to encode and decode luminance in images:

- Compensates for the nonlinear response of display devices
- Mimics the nonlinear perception of brightness in human vision
- Typically applies a power-law function: $C_{out} = C_{in}^{\gamma}$
- Improves the perceived image quality, especially in darker regions

## Color Models

**Question 17.** Explain the RGB color model and how colors are represented in this model.

**Answer:** The RGB color model is an additive color model:

- Uses three primary colors: Red, Green, and Blue
- Colors are represented as triplets (R, G, B)
- Each component typically ranges from 0 to 255 (8-bit depth)
- (0, 0, 0) represents black, (255, 255, 255) represents white
- Mixing different intensities produces a wide range of colors
- Widely used in digital displays and image representation
- Mimics human visual perception

**Question 18.** Explain the YCbCr color model. How does it separate luminance from chrominance information?

**Answer:** The YCbCr color model separates luminance and chrominance:

- Y: Luminance (brightness) component
- Cb: Blue-difference chroma component
- Cr: Red-difference chroma component

Separation allows for efficient compression by allocating more bandwidth to luminance (Y) channel, which human vision is more sensitive to, and less to chrominance (Cb, Cr) channels.

**Question 19.** Convert the following RGB values to YCbCr: R = 150, G = 75, B = 0

**Answer:** Using the standard formula for 8 bit to 8 bit conversion, using the full range $[0, 255]$, we get

$$Y = 0.299 \cdot 150 + 0.587 \cdot 75 \approx 89 \quad (1)$$
$$Cb = -0.168736 \cdot 150 - 0.331264 \cdot 75 + 128 \approx 78 \quad (2)$$
$$Cr = 0.5 \cdot 150 - 0.418688 \cdot 75 + 128 \approx 172. \quad (3)$$

The results were rounded to the nearest integer.

**Question 20.** Given the following YCbCr values, convert them back to RGB: Y = 100, Cb = 50, Cr = 25

**Answer:** Using the inverse of the conversion above, we get

$$\begin{aligned}
\begin{bmatrix} R \\ G \\ B \end{bmatrix} &\approx \begin{bmatrix} 1.00000000 & -0.000926 & 1.401687 \\ 1.00000000 & -0.343695 & -0.714169 \\ 1.00000000 & 1.772160 & 0.000990 \end{bmatrix} \left( \begin{bmatrix} Y \\ Cb \\ Cr \end{bmatrix} - \begin{bmatrix} 0 \\ 128 \\ 128 \end{bmatrix} \right) \\
&\approx \begin{bmatrix} 1.00000000 & -0.000926 & 1.401687 \\ 1.00000000 & -0.343695 & -0.714169 \\ 1.00000000 & 1.772160 & 0.000990 \end{bmatrix} \begin{bmatrix} 100 \\ -78 \\ -103 \end{bmatrix} \\
&\approx \begin{bmatrix} -44 \\ 200 \\ -38 \end{bmatrix}.
\end{aligned}$$

After clamping to $[0, 255]$, this leads to $R = 0$, $G = 200$, $B = 0$.

**Question 21.** Explain the difference between additive and subtractive color models. Provide examples of each.

**Answer:** Additive vs. Subtractive Color Models:

- Additive: Colors are created by adding light of different wavelengths
  - Example: RGB (Red, Green, Blue) used in digital displays
  - Primary colors combine to create white
- Subtractive: Colors are created by subtracting wavelengths from white light
  - Example: CMYK (Cyan, Magenta, Yellow, Key/Black) used in printing
  - Primary colors combine to create black

## Image as a Function

**Question 22.** Define an image as a two-dimensional function.

**Answer:** An image can be defined as a two-dimensional function $f(x, y)$ where:

- $(x, y)$ are spatial coordinates in the image plane
- $f$ is the intensity or gray level at any point $(x, y)$
- For color images, $f$ is a vector of color component values

This function maps each pixel location to its corresponding intensity or color value.

**Question 23.** An image is represented as a matrix of pixel values. For a 3x3 grayscale image, given by the matrix:

$$I = \begin{bmatrix} 50 & 75 & 100 \\ 125 & 150 & 175 \\ 200 & 225 & 250 \end{bmatrix}$$

Apply a linear transformation that scales the pixel values by a factor of 0.8.

**Answer:** Applying the linear transformation $I' = 0.8 \cdot I$:

$$I' = 0.8 \cdot \begin{bmatrix} 50 & 75 & 100 \\ 125 & 150 & 175 \\ 200 & 225 & 250 \end{bmatrix} = \begin{bmatrix} 40 & 60 & 80 \\ 100 & 120 & 140 \\ 160 & 180 & 200 \end{bmatrix}$$

The resulting image has reduced intensity values.

**Question 24.** For a grayscale image of size 1024x768 with 8-bit depth, calculate:

a) The total number of bits required to store the image

b) The number of distinct gray levels that can be represented

**Answer:** a) Total number of bits:

$$\begin{aligned} \text{Bits} &= \text{Width} \times \text{Height} \times \text{Bit Depth} \\ &= 1024 \times 768 \times 8 \\ &= 6,291,456 \text{ bits} \\ &= 786,432 \text{ bytes} \\ &\approx 768 \text{ KB} \end{aligned}$$

b) Number of distinct gray levels:

$$\begin{aligned} \text{Gray Levels} &= 2^{\text{Bit Depth}} \\ &= 2^8 = 256 \end{aligned}$$

An 8-bit grayscale image can represent 256 distinct shades of gray, from 0 (black) to 255 (white).

**Question 25.** Explain the concept of dithering in image processing. How does it trade spatial resolution for increased perceived color depth?

**Answer:** Dithering is a technique used to create the illusion of color depth in images with a limited color palette:

- Approximates unavailable colors by using patterns of available colors
- Trades spatial resolution for perceived color depth
- Exploits the eye's tendency to blend colors over small areas

By sacrificing some spatial detail, dithering can create the perception of a wider range of colors or shades than are actually available in the color palette.

## Sampling and Quantization

**Question 26.** Explain the concepts of sampling and quantization in digital imaging.

**Answer:** In digital imaging:

- Sampling:
  - Discretization of the spatial domain
  - Converts continuous image to a grid of pixels

  - Determined by spatial resolution (e.g., 1920x1080)

- Quantization:
  - Discretization of the intensity/color values
  - Assigns discrete levels to continuous intensity range
  - Determined by bit depth (e.g., 8-bit, 16-bit)

Together, sampling and quantization convert a continuous image into a discrete digital representation.

**Question 27.** Given a continuous image function, describe the steps required to convert it to a discrete digital image.

**Answer:** Steps to convert a continuous image to a discrete digital image:

1. Sampling:
   - Define a sampling grid (e.g., 512x512 pixels)
   - Sample the continuous function at grid points
2. Quantization:
   - Choose bit depth (e.g., 8-bit for 256 levels)
   - Map continuous intensity values to discrete levels
3. Discretization:
   - Assign quantized values to each sampled point
4. Storage:
   - Store the discrete values in a digital format

**Question 28.** What are the effects of decreasing the bit depth of an image from 8 bits to 4 bits?

**Answer:** Effects of reducing bit depth from 8 to 4 bits:

- Decrease in number of possible intensity levels (256 to 16)
- Increased quantization error
- Visible banding in smooth gradients
- Potential loss of fine details and texture
- Reduced file size
- Reduction in image-quality

**Question 29.** Compute the number of quantization levels for a 6-bit grayscale image.

**Answer:** For a 6-bit grayscale image, we have $2^6 = 64$ distinct shades of gray.

# Discrete Cosine Transform (DCT)

**Question 30.** Write the 2D DCT formula for an 8x8 block of image pixels.

**Answer:** The 2D DCT formula for an 8x8 block of image pixels $f(x, y)$ is:

$$G(i,v) = \frac{1}{2}C(v)\sum_{j=0}^{7}\cos\left(\frac{(2j+1)v\pi}{16}\right)f(i,j)$$

$$F(u,v) = \frac{1}{2}C(u)\sum_{i=0}^{7}\cos\left(\frac{(2i+1)u\pi}{16}\right)G(i,v)$$

Where:

- $F(u, v)$ is the DCT coefficient at coordinates $(u, v)$
- $f(x, y)$ is the pixel value at coordinates $(x, y)$
- $C(w) = \frac{1}{\sqrt{2}}$ for $w = 0$, and 1 otherwise

**Question 31.** For the following 2x2 image block, compute the DCT:

$$I = \begin{bmatrix} 100 & 120 \\ 130 & 150 \end{bmatrix}$$

**Answer:** For a 2x2 block, we use:

$$G(i,v) = \frac{1}{2}C(v)\sum_{j=0}^{1}c(j,v)f(i,j)$$

$$F(u,v) = \frac{1}{2}C(u)\sum_{i=0}^{1}c(i,u)G(i,v)$$

with $c(i,u) = \cos\left[\frac{(2i+1)u\pi}{4}\right]$. Note that $c(0,0) = c(1,0) = 1$, $c(0,1) = \sqrt{2}/2$ and $c(1,1) = -\sqrt{2}/2$. This leads us to

$$\begin{aligned}
G(0,0) &= \frac{1}{2}\cdot\frac{\sqrt{2}}{2}(1\cdot 100 + 1\cdot 120) && = \sqrt{2}\cdot 55 \\
G(1,0) &= \frac{1}{2}\cdot\frac{\sqrt{2}}{2}(1\cdot 130 + 1\cdot 150) && = \sqrt{2}\cdot 70 \\
G(0,1) &= \frac{1}{2}\cdot 1(\frac{\sqrt{2}}{2}\cdot 100 + \frac{-\sqrt{2}}{2}\cdot 120) && = -\sqrt{2}\cdot 5 \\
G(1,1) &= \frac{1}{2}\cdot 1(\frac{\sqrt{2}}{2}\cdot 130 + \frac{-\sqrt{2}}{2}\cdot 150) && = -\sqrt{2}\cdot 5
\end{aligned}$$

and

$$\begin{aligned}
F(0,0) &= \frac{1}{2}\cdot\frac{\sqrt{2}}{2}(1\cdot\sqrt{2}\cdot 55 + 1\cdot\sqrt{2}\cdot 70) && = \frac{125}{2} \\
F(1,0) &= \frac{1}{2}\cdot 1(\frac{\sqrt{2}}{2}\cdot\sqrt{2}\cdot 55 + \frac{-\sqrt{2}}{2}\cdot(-\sqrt{2}\cdot 5)) && = 30 \\
F(0,1) &= \frac{1}{2}\cdot\frac{\sqrt{2}}{2}(1\cdot(-\sqrt{2}\cdot 5) + 1\cdot(-\sqrt{2}\cdot 5)) && = -5 \\
F(1,1) &= \frac{1}{2}\cdot 1(\frac{\sqrt{2}}{2}\cdot(-\sqrt{2}\cdot 5) + \frac{-\sqrt{2}}{2}\cdot(-\sqrt{2}\cdot 5)) && = 0
\end{aligned}$$

The resulting DCT coefficients are:

$$F = \begin{bmatrix} \frac{125}{2} & -5 \\ 30 & 0 \end{bmatrix}$$

**Question 32.** Explain why the DCT is preferred over the Fourier Transform for image compression in formats like JPEG.

**Answer:** The DCT is preferred over the Fourier Transform for image compression because:

- It produces real-valued coefficients (FT produces complex values)
- Energy compaction: The DCT has superior energy compaction properties for typical image data. This means it can concentrate most of the signal energy into a few low-frequency components. For natural images, which tend to have mostly low-frequency content, this allows for more efficient compression.
- Faster computation: can be implemented with fewer operations
- Closer to human visual perception of spatial frequencies
- Allows for easier separation of luminance and chrominance information

These properties make DCT more efficient and effective for image compression in formats like JPEG.

## Fourier Transform in Image Processing

**Question 33.** Write the formula for the 2D discrete Fourier transform (DFT).

**Answer:** The 2D Discrete Fourier Transform (DFT) for an image $f(x, y)$ of size $M \times N$ is given by:

$$F(u, v) = \frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi(\frac{ux}{M} + \frac{vy}{N})}$$

Where:

- $F(u, v)$ is the Fourier coefficient at frequencies $(u, v)$
- $f(x, y)$ is the image pixel value at coordinates $(x, y)$
- $M$ and $N$ are the image dimensions
- $j$ is the imaginary unit ($\sqrt{-1}$)

**Question 34.** Explain the relationship between the amplitude spectrum and the phase spectrum of an image's Fourier transform.

**Answer:** The relationship between amplitude and phase spectra:

- Amplitude spectrum:
  - Represents the magnitude of each frequency component

  - Indicates the strength of various frequencies in the image
  - Relates to the contrast and overall structure of the image
- Phase spectrum:
  - Represents the phase angle of each frequency component
  - Encodes the relative positions of different frequency components
  - Crucial for preserving edge information and fine details
- Relationship:
  - Both are needed to fully reconstruct the original image
  - Amplitude affects overall appearance, phase affects structure

**Question 35.** Consider an image with vertical stripes. Describe how the stripes would appear in the frequency domain after applying the Fourier transform.

**Answer:** In the frequency domain, vertical stripes would appear as:

- Bright spots along the horizontal axis of the amplitude spectrum
- The distance of these spots from the origin corresponds to the frequency of the stripes
- Closer spots to the origin indicate lower frequency (wider stripes)
- Spots further from the origin indicate higher frequency (narrower stripes)
- The intensity of the spots relates to the contrast of the stripes

**Question 36.** Compute the 2D Fourier transform for the following 2x2 image block:

$$I = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

**Answer:** For a 2x2 image, we use the 2D DFT formula

$$F(u,v) = \frac{1}{4}\sum_{x=0}^{1}\sum_{y=0}^{1} f(x,y)e^{-j2\pi(\frac{ux}{2}+\frac{vy}{2})}.$$

Let us define $h(ux, vy) = e^{-j2\pi(\frac{ux}{2}+\frac{vy}{2})}$. Notice that $h(0,0) = h(1,1) = 1$ and $h(1,0) = h(0,1) = -1$, we can compute

$$\begin{aligned} F(0,0) &= \frac{1}{4}(1+0+0+(-1)) &&= 0 \\ F(0,1) &= \frac{1}{4}(1-(-1)) &&= \frac{1}{2} \\ F(1,0) &= \frac{1}{4}(1-(-1)) &&= \frac{1}{2} \\ F(1,1) &= \frac{1}{4}(1+(-1)) &&= 0 \end{aligned}$$

The resulting Fourier transform is:

$$F = \begin{bmatrix} 0 & \frac{1}{2} \\ \frac{1}{2} & 0 \end{bmatrix}$$

**Question 37.** Explain how to reconstruct an image from its Fourier transform.

**Answer:** To reconstruct an image from its Fourier transform:

1. Apply the Inverse Fourier Transform (IFT):

$$f(x,y) = \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F(u,v) e^{j2\pi(\frac{ux}{M} + \frac{vy}{N})}$$

2. Ensure both magnitude and phase information are preserved
3. Handle any numerical precision issues (round-off errors)
4. Normalize the result to fit the original image's intensity range
5. For real images, take the real part of the complex result
6. Apply any necessary post-processing (e.g., clipping values)

This process reverses the Fourier transform, converting frequency domain information back to the spatial domain.

**Question 38.** If an image's high-frequency components are removed, what changes would you expect in the reconstructed image?

**Answer:** Removing high-frequency components would result in:

- Blurring of the image
- Loss of fine details and textures
- Softening of edges
- Reduction in overall sharpness
- Smoothing of rapid intensity changes
- Potential loss of noise (if present in the original)
- Preservation of overall shape and low-frequency content

The image would appear smoother but with less detail, as high frequencies correspond to rapid spatial variations in intensity.

## Lossy Compression

**Question 39.** Describe the process of quantizing DCT coefficients in JPEG compression.

**Answer:** The process of quantizing DCT coefficients in JPEG compression:

1. Divide the image into 8x8 blocks
2. Apply DCT to each block
3. For each block, divide DCT coefficients by a quantization matrix:

$$F_Q(u,v) = \text{round}\left(\frac{F(u,v)}{Q(u,v)}\right)$$

4. The quantization matrix $Q$ is designed to preserve low frequencies and reduce high frequencies
5. Adjust $Q$ based on desired compression level (higher compression = larger values in $Q$)
6. Round the result to the nearest integer
7. Many high-frequency coefficients become zero, enabling compression

This process introduces controlled loss of information, trading image quality for reduced file size.

**Question 40.** After applying quantization, which DCT coefficients are typically most affected, and why?

**Answer:** After quantization:

- Most affected: High-frequency coefficients
  - Correspond to fine details and rapid changes in the image
  - Human vision is less sensitive to these frequencies
  - Quantization matrix has larger values for high frequencies
- Least affected: Low-frequency coefficients
  - Represent overall structure and smooth variations
  - Human vision is more sensitive to these frequencies
  - Quantization matrix has smaller values for low frequencies

This selective quantization allows for significant compression while maintaining perceptual quality.

**Question 41.** Apply a basic quantization matrix to the DCT coefficients computed in Question 31 and analyze the result.

**Answer:** Let's use a simple quantization matrix for our 2x2 DCT result:

$$Q = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$$

Applying quantization to our DCT coefficients:

$$F = \begin{bmatrix} \frac{125}{2} & -5 \\ 30 & 0 \end{bmatrix}$$

Quantized result:

$$F_Q = \begin{bmatrix} \text{round}(125/2) & \text{round}(-10\sqrt{2}/2) \\ \text{round}(30/2) & \text{round}(0/4) \end{bmatrix} = \begin{bmatrix} 63 & -2 \\ 15 & 0 \end{bmatrix}$$

Analysis:

- DC coefficient (top-left) is preserved
- High-frequency coefficient (bottom-right) is still zero
- Mid-frequency coefficients are rounded and reduced
- Some information loss, but major features are retained

**Question 42.** What is the main trade-off between lossy and lossless image compression?

**Answer:** The main trade-off between lossy and lossless compression:

- Lossy compression:
  - Pros: Higher compression ratios, smaller file sizes
  - Cons: Irreversible loss of information, potential quality degradation
- Lossless compression:
  - Pros: Perfect reconstruction of original image, no quality loss
  - Cons: Lower compression ratios, larger file sizes

The choice depends on the application: lossy for general-purpose images where some quality loss is acceptable, lossless for medical, scientific, or archival purposes where data integrity is crucial.

**Question 43.** JPEG is a widely used lossy compression format. Describe how JPEG compresses an image using DCT and quantization.

**Answer:** JPEG compression process:

1. Color space conversion: RGB to YCbCr
2. Chroma subsampling (optional)
3. Divide image into 8x8 blocks
4. Apply DCT to each block
5. Quantization of DCT coefficients:
   - Use standardized or custom quantization matrices
   - Divide DCT coefficients by quantization values and round
6. Zigzag ordering of quantized coefficients
7. Run-length encoding of zero coefficients
8. Entropy coding (typically Huffman coding)

This process exploits spatial redundancy, psychovisual redundancy, and coding redundancy to achieve compression while maintaining acceptable visual quality.

## Lossless Compression

**Question 44.** Explain how PNG achieves lossless compression.

**Answer:** PNG (Portable Network Graphics) achieves lossless compression through:

1. Pass extraction
2. Scanline serialization
3. Filtering
4. Compression
5. Chunking
6. Datastream construction

This approach allows for perfect reconstruction of the original image while still achieving significant compression.

**Question 45.** Compare the compression ratios of JPEG and PNG for the same image. What accounts for the difference?

**Answer:** Comparison of JPEG and PNG compression ratios:

- JPEG typically achieves higher compression ratios
- PNG generally results in larger file sizes

Factors accounting for the difference:

- Lossy vs. Lossless:
  - JPEG: Lossy compression allows for higher ratios
  - PNG: Lossless compression preserves all data
- Compression techniques:
  - JPEG: DCT and quantization exploit psychovisual redundancy
  - PNG: focuses on statistical redundancy
- Image content:
  - JPEG: Efficient for natural images with smooth color transitions
  - PNG: Better for images with sharp edges, text, or graphics
- Color depth:
  - JPEG: Limited to 8 bits per channel
  - PNG: Supports higher bit depths, potentially increasing file size

The choice between JPEG and PNG depends on the specific image content and the requirements for quality vs. file size.

**Question 46.** Describe the principle of predictive coding in lossless JPEG compression.

**Answer:** Predictive coding in lossless JPEG:

1. Prediction: Estimate pixel values based on neighboring pixels
    - Uses one of several predefined predictors
    - E.g., left neighbor, top neighbor, or average of both
2. Difference calculation: Compute the difference between actual and predicted values
3. Entropy coding: Apply Huffman coding to the differences
    - Smaller differences are more frequent and get shorter codes
4. Lossless reconstruction: Add predicted values to decoded differences

This approach exploits spatial correlations in the image, reducing redundancy without loss of information.

## Elementary Image Expansion

**Question 47.** Explain how an image can be expressed as a linear combination of elementary images using the outer product of vectors.

**Answer:** An image can be expressed as a linear combination of elementary images using outer products:

- Let $I$ be an $N \times N$ image matrix
- Express $I$ as a sum:
$$I = \sum_{i=1}^{N} \sum_{j=1}^{N} g_{i,j} u_i v_j^T$$
where $u_i, v_j^T$ determine the specific image representation.
- The sum reconstructs the original image
- This decomposition can be achieved through Singular Value Decomposition (SVD)
- or other approaches

This representation allows for image compression and analysis by focusing on the most significant components.

**Question 48.** Expand the following 2x2 image into elementary images:

$$I = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

**Answer:** To expand this 2x2 image into elementary images, we'll use Singular Value Decomposition (SVD):
$I = U\Sigma V^T$
Calculating SVD (approximated values):

$$U \approx \begin{bmatrix} 0.4046 & 0.9145 \\ 0.9145 & -0.4046 \end{bmatrix}$$
$$\Sigma \approx \begin{bmatrix} 5.4650 & 0 \\ 0 & 0.3660 \end{bmatrix}$$
$$V^T \approx \begin{bmatrix} 0.5760 & 0.8174 \\ 0.8174 & -0.5760 \end{bmatrix}$$

Now, we can express $I$ as the sum of two elementary images:

$$\begin{aligned} I &= \sigma_1 u_1 v_1^T + \sigma_2 u_2 v_2^T \\ &\approx 5.4650 \begin{bmatrix} 0.4046 \\ 0.9145 \end{bmatrix} \begin{bmatrix} 0.5760 & 0.8174 \end{bmatrix} \\ &+ 0.3660 \begin{bmatrix} 0.9145 \\ -0.4046 \end{bmatrix} \begin{bmatrix} 0.8174 & -0.5760 \end{bmatrix} \\ &\approx \begin{bmatrix} 0.9748 & 1.9434 \\ 2.9434 & 3.9121 \end{bmatrix} + \begin{bmatrix} 0.0252 & 0.0566 \\ 0.0566 & 0.0879 \end{bmatrix} \end{aligned}$$

The sum of these two elementary images reconstructs the original image $I$. Note that this only has two no-zero elementary images, as $\Sigma$ only has two no-zero entries. In our above notation, we would write $g_{i,j} = \Sigma_{i,j}$.

## Basis Transformations

**Question 49.** What are the advantages of using the DCT for image compression compared to using the Fourier transform?

**Answer:** Advantages of DCT over Fourier transform for image compression:

- Real-valued coefficients (FT produces complex values)
- Better energy compaction in fewer coefficients
- Faster computation (fewer operations required)
- Reduced "block boundary" artifacts in block-based compression
- Closer alignment with human visual perception of spatial frequencies
- Higher computational efficiency

**Question 50.** Apply the Fourier transform to a small 4x4 image and compare the result with applying the DCT.

**Answer:** Let's use a simple 4x4 image:

$$g = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}$$

The unitary matrix ix

$$U_{x\alpha} = \frac{1}{2} e^{-j\frac{\pi x\alpha}{2}}$$

$$U = \frac{1}{2}\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix}$$

this leads to

$$\hat{g} = UgU$$

$$= \frac{1}{4}\begin{bmatrix} 10 & 1+j & 4 & 1-j \\ -2 & 1+j & 0 & 1-j \\ 2 & 1+j & -4 & 1-j \\ -2 & 1+j & 0 & 1-j. \end{bmatrix}$$

Now let us form the DCT: The unitary matrix is

$$U^{DCT} \approx \begin{bmatrix} 0.5 & 0.5 & 0.5 & 0.5 \\ 0.65 & 0.27 & -0.27 & -0.65 \\ 0.5 & 0.5 & 0.5 & 0.5 \\ 0.27 & -0.65 & 0.65 & -0.27, \end{bmatrix}$$

leading to

$$UgU^T \approx \begin{bmatrix} 2.50 & 0.38 & 0.05 & 0.92 \\ 0.00 & 1.00 & 0.00 & 0.00 \\ 0.00 & 0.00 & 1.00 & 0.00 \\ 0.00 & 0.00 & 0.00 & 1.00 \end{bmatrix}$$

Comparison:

- DFT produces complex values, DCT produces real values
- DCT concentrates more energy in fewer coefficients (top-left)
- DCT has a simpler structure, easier to compress
- DCT coefficients directly relate to visual features (e.g., DC term is average brightness)

## Spatial Filtering

**Question 51.** Describe how spatial filtering can be used to enhance edges in an image.

**Answer:** Spatial filtering enhances edges in an image by emphasizing pixel intensity differences at the boundaries of objects. This is typically done with convolution kernels designed to highlight regions where pixel intensity changes rapidly. Here's how it works:

1. Use edge detection filter: Filters like the Sobel, Prewitt, and Laplacian are commonly used. These kernels detect gradients, or intensity changes, by calculating differences between neighboring pixel values. The result is a new image highlighting the edges where changes are most significant.

2. Convolve the kernel with the image: The chosen kernel is passed over the entire image, convolving each region. For edge enhancement, the kernel amplifies regions with high-intensity variation, outputting bright pixels along edges and leaving low-gradient areas (uniform regions) with minimal change.

3. Sharpening with High-Pass Filters: High-pass filters (e.g., Laplacian) can enhance edges further by adding the result back to the original image. This boosts high-frequency details, making edges appear sharper and more defined.

These methods increase the contrast between edges and their surroundings, making them more prominent.

**Question 52.** Write the 3x3 convolution kernel for a simple edge-detection filter. Apply this filter to a 3x3 image block:

$$I = \begin{bmatrix} 50 & 75 & 100 \\ 125 & 150 & 175 \\ 200 & 225 & 250 \end{bmatrix}$$

**Answer:** A simple 3x3 edge detection kernel (Laplacian):

$$K = \begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

Applying the filter to the image block (using symmetrical padding):

$$I' = \begin{bmatrix} -100 & -75 & -50 \\ -25 & 0 & 25 \\ 50 & 75 & 100 \end{bmatrix}$$

## Smoothing and Sharpening

**Question 53.** Explain the difference between smoothing and sharpening filters. Provide examples of each.

**Answer:** Smoothing vs. Sharpening filters:
Smoothing filters:

- Reduce noise and fine details
- Blur the image
- Low-pass filters in frequency domain
- Example:
  - Mean filter: $\frac{1}{9}\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$
  - Gaussian filter: $\frac{1}{16}\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$

Sharpening filters:

- Enhance edges and fine details
- Increase contrast between adjacent areas
- High-pass filters in frequency domain
- Example:
  - Laplacian filter: $\begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$

Smoothing reduces high-frequency components, while sharpening enhances them.

**Question 54.** Given a 3x3 Gaussian smoothing filter,

$$G = \frac{1}{16} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

convolve it with the following image block:

$$I = \begin{bmatrix} 100 & 120 & 130 \\ 110 & 115 & 125 \\ 140 & 150 & 160 \end{bmatrix}$$

**Answer:** To convolve the Gaussian filter with the image block, we multiply corresponding elements using symmetrical padding:

$$I' \approx \begin{bmatrix} 107 & 117 & 126 \\ 118 & 125 & 133 \\ 135 & 142 & 149 \end{bmatrix}$$

**Question 55.** Describe how the concepts of spatial frequency and the Nyquist sampling theorem extend to 2D image sampling. What are the implications for avoiding aliasing in digital images?

**Answer:** Spatial frequency and Nyquist theorem in 2D image sampling:

- Spatial frequency: Rate of change of intensity in x and y directions
- 2D Nyquist theorem: Sampling rate must be at least twice the highest spatial frequency in both dimensions
- Implications for avoiding aliasing:
  - Sample at a rate higher than twice the highest frequency component
  - Use anti-aliasing filters before sampling to remove high frequencies
  - Choose appropriate sensor resolution for the desired image detail
  - Implement proper reconstruction filters when resizing images
- Consequences of undersampling:

  - Moiré patterns in periodic structures
  - Jagged edges (jaggies) in diagonal lines
  - Loss of fine details and textures

- Techniques to reduce aliasing:
  - Optical low-pass filters in cameras
  - Supersampling and downsampling in computer graphics
  - Adaptive sampling based on local frequency content

Understanding these concepts helps in designing imaging systems and processing algorithms that maintain image quality and fidelity.